Small road-damage detection

Across unseen countries

Three controlled studies in assignment, consistency and missed-damage prediction

A **ASSIGN**

B **STABILISE**

C **FLAG MISSES**

Three-member final-year research project

The problem: country shift and silent misses

TRAINING CONTEXT

Source-country roads



UNSEEN CONTEXT

Different roads + cameras

Appearance changes

Surface, lighting, markings and viewpoint shift.

Small / thin damage

Its recall under shift must be measured.

Misses stay invisible

A missing box gives an operator no warning.

Research aim: measure transfer and flag likely misses without target-country data.

Sources: proposal v8, pp. 2-3; Arya et al. (2020); RDD2022 / CRDDC'2022

The gap is in the controlled comparisons

Source-only road-damage transfer already exists: MAMDG (Chen et al., 2026).

	EXISTING IDEA	OUR QUESTION ON AN UNSEEN COUNTRY
A	**Size-aware matching / NWD**	Does short-side scaling beat equally tuned plain NWD?
B	**Consistency across views**	Does choosing small / thin boxes beat generic consistency?
C	**Prediction of detector misses**	Do features add value beyond learned weak detections?

Bounded to papers examined. Full MAMDG experiments and MSDA-RDD reading remain a week 1-3 gate.

Sources: Chen et al. (2026), MAMDG; Wang et al. (2021), NWD; Danish et al. (2024); Rahman et al. (2019)

Four folds, with development kept source-only

	Japan	India	United States	Czech Republic
Fold 1	HELD OUT	SOURCE	SOURCE	SOURCE
Fold 2	SOURCE	HELD OUT	SOURCE	SOURCE
Fold 3	SOURCE	SOURCE	HELD OUT	SOURCE
Fold 4	SOURCE	SOURCE	SOURCE	HELD OUT

1 Develop → **2 Freeze** → **3 Retrain** → **4 Evaluate**

2 countries + inner validation

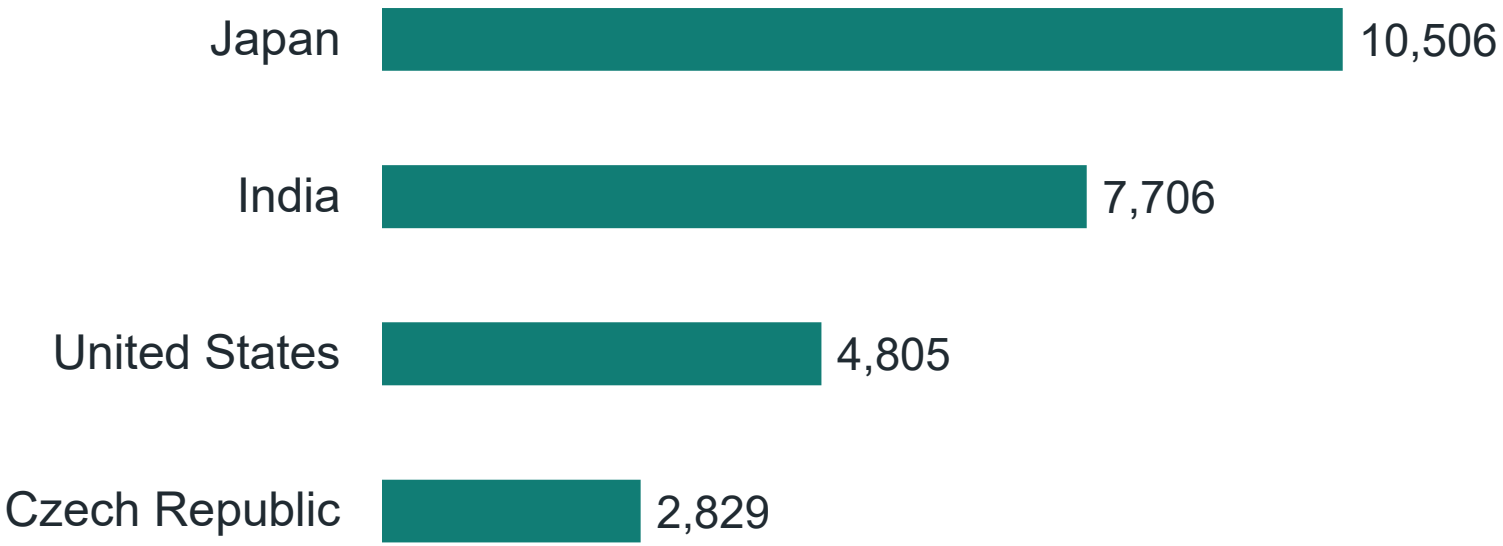
Rules and code fixed

All 3 source countries

Open the held-out country once

Source: proposal v8, pp. 9-11 | Baseline: YOLO11s, 640 px, COCO weights; Ultralytics 8.4.192

RDD2022: 25,846 labelled images



BOX GEOMETRY AT 640 PX



Small

Area < 32 × 32 px



Thin

Short side < 16 px

D00 longitudinal / D10 transverse / D20 alligator / D40 pothole

Keep duplicate groups together. Keep labelled negatives. Flag rare classes. Box size is not crack severity.

Source: Arya et al. (2024), RDD2022, Geoscience Data Journal; proposal v8, pp. 8-9 | CC BY-SA 4.0

A: scale assignment similarity by the short side

PLAIN NWD

$$S = \exp(-W_2 / C)$$

C is constant for all boxes

SHORT-SIDE-RELATIVE NWD

$$S = \exp(-W_2 / (\kappa \times s))$$

s = min(width, height) of the labelled box

SAME 2 PX DISTANCE, K = 1



2 px short side → S ≈ 0.37



20 px → S ≈ 0.90

Stricter on thin boxes; affects assignment and target-score weights.

Sources: proposal v8, pp. 12-13; Wang et al. (2021), NWD | Values below are a mathematical illustration

A: test normalisation against a fair rival

PROPOSED METHOD

Short-side-relative NWD

vs

PRINCIPAL COMPARATOR

Plain NWD with equally tuned C

Keep fixed

Candidate eligibility, data, training schedule and tuning budget.

Primary outcome

Held-out mAP50:95, averaged equally across countries.

Explain the mechanism

Positives per box, zero-positive boxes and target-weight mass.

H1: higher held-out mAP50:95. Small / thin breakdowns are secondary; noisy boxes are extension-only.

Source: proposal v8, pp. 6, 12-13, 16-17 | Original training labels in the core study

B: anchor consistency on small and thin boxes



Bernoulli KL on class scores + L1 on normalised boxes

Test only the 640 px student. Exclude boxes with short side < 8 px at full size; report their share.

Sources: proposal v8, pp. 13-14; Danish et al. (2024), CVPR | Proposed training graph

B: separate object choice from extra training

PROPOSED METHOD

Consistency on small / thin boxes

vs

PRINCIPAL COMPARATOR

Consistency on all eligible boxes

Matched training

Identical views, supervision, loss and training budget.

Primary outcome

Small-or-thin recall at a source-chosen threshold.

Secondary control

Random boxes: same count as the anchored set; one seed.

H2 requires higher recall AND false positives per image ≤ 1.10 × generic consistency.

Source: proposal v8, pp. 7, 13-14, 17 | Thresholds are chosen on source data for each final detector

C: predict where the detector missed damage

INPUTS FROM A FROZEN DETECTOR

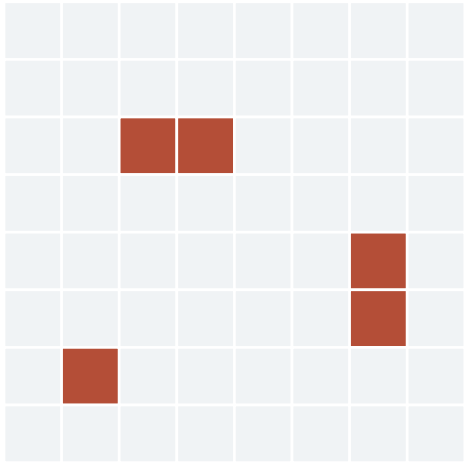
P3 + P4 features

Weak-detection map

Scores from 0.05 to the source-chosen threshold



40 × 40 MISS-LOCATION GRID



Schematic grid detail

Positive cell = centre of an unmatched small / thin labelled box

Same-class, one-to-one matching at IoU ≥ 0.5. Flags likely locations; does not recover missing cracks.

Sources: proposal v8, pp. 14-15; Rahman et al. (2019), IROS; Zimmermann et al. (2026), KGFP

C: ask whether internal features add information

PROPOSED METHOD

Head on features + weak map

vs

PRINCIPAL COMPARATOR

Matched learned head on weak map only

Primary outcome

Cell-level AUPRC, with the positive-cell share stated.

Additional baselines

Raw weak map and a features-only head.

No detector leakage

Separate source slices fit and select the head.

Week 8 gate: if the grid target is unusable, switch to an image-level miss predictor.

Source: proposal v8, pp. 7, 14-15, 17 | C is evaluated first on the shared baseline detector

Evaluate each claim on the same held-out countries

STUDY	PRINCIPAL CONTRAST	PRIMARY OUTCOME
A	Short-side NWD vs plain NWD	**mAP50:95**
B	Anchored vs generic consistency	**Small / thin recall**
C	Features + weak map vs weak map only	**Cell-level AUPRC**

Report variation

Each country, country-macro mean and worst country.

Paired uncertainty

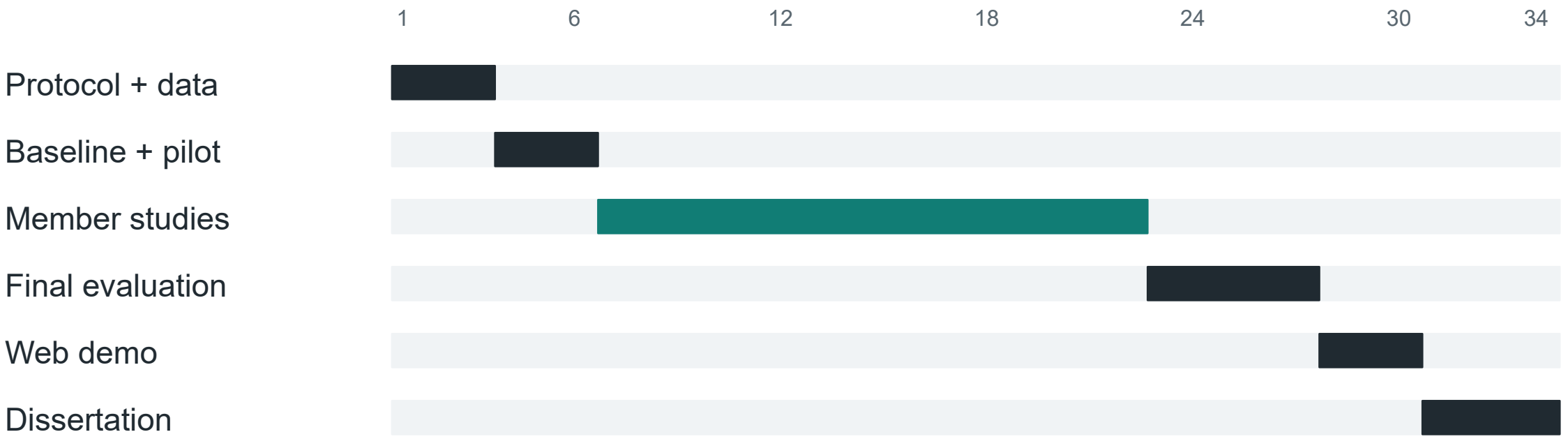
Resample images / duplicate groups; average seed-wise differences.

Bound the claim

Four countries do not establish worldwide generalisation.

Source: proposal v8, pp. 16-18 | Three seeds for principal contrasts; one-seed controls are exploratory

Three individual studies, one 34-week plan



Gates: W3 literature / protocol · W6 GPU timing · W8 C target viability · W22 global freeze

A: assignment + logs B: consistency + recall C: miss prediction Shared: data, evaluation and demo

Source: proposal v8, pp. 21-23, step-by-step method pp. 34-39 | Weeks are relative to project approval

Feasibility depends on the week-6 GPU pilot

72

final detector trainings

AB and noise study excluded.
Plus 20-30 development runs.

500 GPU-hours

If committed, with 100 reserved for development:

ordinary run must be ≈ 3.5 h or less

86 run equivalents × 1.3 allowance × measured run time

Cut exploratory work first; benchmark YOLO11n if needed.

Retain all four countries, three principal-contrast seeds, plain NWD and generic consistency.

Source: proposal v8, pp. 18-23 | GPU times and feature-cache size are planning estimates, not measurements

Expected contribution: three testable answers

A **Short-side-relative assignment**

Does the normalisation improve held-out detection?

B **Small / thin consistency**

Does object selection improve recall at controlled false positives?

C **Miss-location prediction**

Do internal features add information beyond weak outputs?

Deliver: evaluated studies + reproducible benchmark + a small web demonstration.

Four-country evidence; likely-miss warnings, not recovered cracks or demonstrated safety benefits.

Source: proposal v8, pp. 15-18, 24 | Proposed outcomes; no experimental results are claimed